Notice, Interpret, Shape, Govern

A Consolidation Note on Weak-Signal Architecture

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Abstract

This note consolidates the PUTMAN / Spanda paper sequence into a compact architecture statement: a four-stage weak-signal stack, a minimal event state machine, a promotion predicate template, and a short implementation checklist. It does not introduce a new theory, subsystem, product, or domain application. It identifies the shared architectural pattern across the work: weak, noisy, partial, or ambiguous signals may influence bounded attention and response before they are allowed to harden into durable memory, authority, escalation, or action.

The core pattern is:

Notice → Interpret → Shape → Govern

The deliverable is a narrow specification for boundary discipline: what may be noticed, how candidate meaning is formed, what reversible shaping actions are allowed, and what must be true before stronger consequence is promoted.

1. Why This Note Exists

The PUTMAN / Spanda work spread across multiple papers, companion notes, bridge papers, demos, and interface specifications. That spread was useful while the architecture was being explored, but it also created a legibility problem. The project can appear to branch into many unrelated directions.

This note exists to make the center explicit.

The center is not any single domain. It is a stratified weak-signal architecture for disciplined interpretation and promotion control. The repeated problem is simple: early signals are often too weak for certainty but too structured for indifference. A useful architecture needs somewhere for those signals to live before they become truth, memory, policy, escalation, or action.

This note is therefore not another bridge paper in the sequence. It is a consolidation note for the sequence.

2. Core Architecture

The architecture separates four jobs that are often collapsed:

bounded finding → candidate meaning → bounded shaping → governed consequence

Or, in shorter form:

Notice → Interpret → Shape → Govern

This is not a claim that weak signals are decisive. It is a way to let weak signals influence short-horizon behavior without granting them premature authority.

2.1 Minimal State Machine

The stack can be expressed as a compact state machine.

Noticed. A bounded finding has been recorded. The system may emit a finding record, source metadata, local feature values, and uncertainty. It may not emit a truth claim, durable memory write, or action authorization.

Interpreted. One or more candidate meanings have been attached to the finding. The system may emit candidate lists, confidence scores, ambiguity status, and provenance links. It may not emit a final belief, baseline update, or irreversible consequence.

Shaped. A short-horizon response or attention adjustment has been queued. The system may emit a reversible action, inspection request, sampling change, clarification prompt, or review band. It may not emit permanent state mutation, policy change, or final promotion.

Governed. A declared promotion decision has been made. The system may promote, deny, decay, quarantine, escalate, commit, or hold for review. It may not produce an ungoverned side effect or unlogged transition.

2.2 Transition Triggers

Noticed → Interpreted. Triggered when a finding arrives with sufficient metadata for local interpretation. Minimum requirement: finding.id, source, timestamp, features, and uncertainty or confidence.

Interpreted → Shaped. Triggered when a candidate meaning warrants bounded short-horizon response. Minimum requirement: a policy rule or confidence/ambiguity condition selects reversible shaping only.

Shaped → Governed. Triggered when the promotion predicate is evaluated. Minimum requirement: persistence, independent support, provenance, audit token, and stability condition are checked.

Any → Decayed / Resolved. Triggered by timeout or return to corridor. Minimum requirement: an expiry rule or decay condition is satisfied.

Any → Quarantined. Triggered when evidence becomes unreliable, contradictory, adversarial, or malformed. Minimum requirement: a quarantine reason and provenance record are emitted.

2.3 Minimal Data Model

A minimal implementation can use four linked records:

```
{
  "finding": {
    "id": "f_001",
    "source": "source_id",
    "timestamp": "ISO-8601",
    "scope": "local_scope",
    "features": {},
    "confidence": 0.0,
    "provenance": []
  },
  "interpretation": {
    "finding_id": "f_001",
    "candidates": [],
    "confidences": [],
    "ambiguity": "low|medium|high",
    "provenance": []
  },
  "shape": {
    "interpretation_id": "i_001",
    "actions": [],
    "reversible": true,
    "expiry": "duration_or_timestamp",
    "scope_limit": "local_only"
  },
  "governance": {
    "decision": "promote|deny|decay|quarantine|review|escalate",
    "promotion_token": "optional",
    "audit_log_id": "audit_001",
    "predicate_vector": {},
    "rationale": []
  }
}
```

The exact schema may vary by domain. The invariant is that finding, interpretation, shaping, and governance remain separately inspectable.

3. Promotion Predicate Template

Governance should not be treated as a magic switch. A candidate may be promoted only when declared predicates are satisfied.

A minimal promotion predicate can be expressed as:

```
PROMOTE(candidate) iff:
  persistence >= P_min
  AND independent_support >= S_min
  AND provenance_chain == intact
  AND audit_token == present
  AND stability_state == stable
  AND contradiction_score < C_max
  AND requested_consequence <= authorized_scope
```

Where:

persistence >= P_min means the candidate lasts or recurs across a declared review window.

independent_support >= S_min means at least the required number of independent sources, cues, layers, or checks support the candidate.

provenance_chain == intact means the evidence path can be replayed or inspected.

audit_token == present means the transition emits a durable audit record.

stability_state == stable means baseline update or stronger consequence is not allowed during unstable regimes.

contradiction_score < C_max means active contradictory evidence remains below a declared blocking threshold.

requested_consequence <= authorized_scope means the action or memory effect stays within granted authority.

Promotion failure does not imply the signal was meaningless. It means the signal has not earned stronger scope.

Possible non-promotion outcomes include:

- keep watching
- request more evidence
- decay
- quarantine
- route to human review
- deny stronger consequence
- preserve as associative trace only

4. Boundary Rules / Anti-Collapse Checklist

The architecture is disciplined by anti-collapse rules. Each rule has an implementation implication.

Noticing is not truth. Finding records may be logged, but they must not update priors or authorize action until promoted.

Salience is not persistence. Reflex or attention signals must expire unless promoted through governed criteria.

Interpretation is not authority. Candidate meanings may shape inspection, but they may not directly commit memory or action.

Shared representation is not shared meaning. Shared tags stabilize reference, not identical uptake across agents or modules.

Association is not commitment. Associative traces may influence attention or recall, but they may not mutate canonical state.

Recall is not authorization. Recalled content must be re-ingested or re-evaluated before promotion.

Promotion requires support and governance. Stronger consequence requires predicates, provenance, stability, and audit.

These rules are the stack's main safety discipline. They prevent weak evidence from hardening too early while still allowing it to affect near-term behavior.

5. Component Map

The main papers define different parts of the same stack.

Bounded Observation. Preserves local, constraint-accessible findings. The relevant artifacts are finding records, evidence spans, and local feature packets. Related papers include CDE, Reflex Attention, and Constraint-Grounded Dimensional Inference.

Provisional Interpretation. Forms candidate meanings under context. The relevant artifacts are candidate interpretations, ambiguity scores, and confidence vectors. Related papers include the PUTMAN Model, Interpretive Deviation and Attention Escalation, and Shared Representation / Divergent Interpretation.

Short-Horizon Salience. Raises attention without persistence or authority. The relevant artifacts are reflex events, micro-baseline scores, and expiry conditions. The main related paper is Reflex Attention.

Typed Representation. Standardizes portable signal exchange. The relevant artifacts are typed tags, dictionary versions, commit classes, and provenance records. Related papers include TVS and S.U.N.

Deviation and Governance. Decides whether deviation becomes structurally relevant. The relevant artifacts are gate levels, review bands, rationale artifacts, and audit records. Related papers include CDE, Normative Interface Contracts, and the Reference Architecture.

Stratified Memory. Controls persistence across temporal layers. The relevant artifacts are short-term, intermediate, and prior-level states, write gates, decay records, and consolidation records. Related papers include Memory Stratification, EEC, and RFM.

Experience Entry Control. Governs the boundary between event and memory. The relevant artifacts are associative traces, canonical state transitions, and transition logs. The main related paper is EEC.

Field / Interaction Dynamics. Models multi-agent coupling and consequence. The relevant artifacts are field updates, reference shift markers, and dissipation or amplification records. Related papers include Multi-Agent Field Dynamics and RFM.

Read together, the papers describe boundary control across observation, interpretation, representation, salience, memory, governance, and interaction.

6. Worked Example: Grid Anomaly Triage

This example is illustrative, not a utility deployment standard. Thresholds are placeholders to show the transfer pattern.

6.1 Input Finding

A monitoring system receives a weak signal:


```
{
  "id": "f_grid_017",
  "source": "PMU_A12",
  "timestamp": "2026-05-12T14:06:00Z",
  "scope": "node_N7 / feeder_F2",
  "features": {
    "single_phase_voltage_deviation": 0.03,
    "snr": "low"
  },
  "confidence": 0.62,
  "provenance": ["pmu_stream_A12", "baseline_24h_N7"]
}
```


The finding is not an incident. It is a bounded deviation record.

6.2 Noticed

The system records:

```
state = Noticed
record = voltage deviation of 3% at node N7
source = PMU_A12
confidence = 0.62
```

Allowed behavior: preserve the finding and compare it against local baseline.

Forbidden behavior: create incident, dispatch response, update long-horizon baseline.

6.3 Interpreted

The interpretation layer attaches candidate meanings:


```
{
  "finding_id": "f_grid_017",
  "candidates": [
    {"label": "transient_load", "confidence": 0.35},
    {"label": "sensor_drift", "confidence": 0.25},
    {"label": "incipient_fault", "confidence": 0.40}
  ],
  "ambiguity": "high",
  "provenance": ["baseline_24h_N7", "recent_feeder_context_F2"]
}
```


The system does not pretend the strongest candidate is settled truth. It preserves competing meanings.

6.4 Shaped

Because the signal is weak but non-trivial, the system applies a reversible shaping action:


```
{
  "actions": [
    "increase_sampling_to_1s",
    "request_SCADA_corrob_window",
    "open_operator_watch_band"
  ],
  "reversible": true,
  "expiry": "10m",
  "scope_limit": "node_N7 / feeder_F2"
}
```


This changes inspection behavior. It does not create durable incident state.

6.5 Governed

Promotion is evaluated using a declared predicate:

```
Promote to operator-facing anomaly if:
  voltage deviation persists across >= 3 consecutive readings
  AND independent SCADA support is present
  AND provenance chain is intact
  AND audit token is emitted
  AND no sensor-drift contradiction exceeds threshold
```

Possible outcomes:

Promote. Persistence and independent support are present; audit record emitted.

Hold / Watch. Deviation persists but independent support is missing.

Decay. Signal returns to the local corridor before the review window closes.

Quarantine. Evidence favors PMU artifact or telemetry fault.

Expected audit record:


```
{
  "decision": "review",
  "candidate": "incipient_fault",
  "scope": "node_N7 / feeder_F2",
  "predicate_vector": {
    "persistence_readings": 2,
    "required_persistence": 3,
    "independent_support": false,
    "provenance_chain": "intact",
    "audit_token": "present",
    "contradiction_score": 0.18
  },
  "rationale": [
    "weak voltage deviation persisted but did not yet meet promotion threshold",
    "SCADA corroboration pending"
  ]
}
```


This is the stack's practical value: the weak signal changes what the system inspects next, but stronger operational significance waits for support.

7. Domain Transfer Examples

The bridge papers demonstrate that the same boundary problem appears across domains. Transfer does not mean identical thresholds or identical costs. Each domain must adapt latency, evidentiary standards, review burden, safety requirements, and regulatory constraints.

Low-altitude drone sensing. Typical weak signals include acoustic irregularity, motion persistence, and weak RF or thermal partials. The promotion cost is avoiding premature track formation or high-impact defensive response.

Grid anomaly triage. Typical weak signals include voltage deviation, telemetry mismatch, thermal irregularity, and DER under-response. The promotion cost is avoiding false operational significance and unnecessary operator load.

Industrial fault detection. Typical weak signals include vibration drift, thermal rise, pressure fluctuation, and acoustic anomaly. The promotion cost is avoiding alarm fatigue or premature maintenance action.

Cyber anomaly triage. Typical weak signals include login context shift, privilege deviation, process ancestry anomaly, and outbound irregularity. The promotion cost is avoiding premature incident state or automated containment.

Robotics anomaly sensing. Typical weak signals include wheel-current asymmetry, micro-slip, pose drift, and force/torque mismatch. The promotion cost is avoiding unsafe motion changes or durable false fault state.

Perimeter anomaly triage. Typical weak signals include motion persistence, thermal inconsistency, route irregularity, and sensor disagreement. The promotion cost is preserving a human-review boundary before stronger incident framing.

Autonomous vehicle edge cases. Typical weak signals include estimator disagreement, perception uncertainty, and scene ambiguity. The promotion cost is avoiding increased irreversible or high-risk actuation under disagreement.

Empathy / frame preservation. Typical weak signals include wording-frame mismatch, relational cue loss, and context omission. The promotion cost is avoiding flattened meaning based only on surface text.

AI safety monitoring. Typical weak signals include runtime anomaly, estimator disagreement, and capability uncertainty. The promotion cost is avoiding direct jumps to shutdown, memory change, or capability escalation.

Healthcare early warning. Typical weak signals include weak physiological trends, contextual irregularities, and repeated borderline cues. The promotion cost is avoiding both missed deterioration and premature clinical conclusion.

The shared architecture is portable only at the boundary level. Domain implementation remains domain-specific.

8. Minimal Test Checklist

A weak-signal implementation should pass at least the following tests.

Test 1 — Salience Does Not Write Memory

Input: a single weak finding crosses a reflex or salience threshold.

Expected result:

- finding is logged
- short-horizon shaping may occur
- no durable memory write occurs
- no baseline update occurs
- expiry is attached

Failure mode caught: salience collapsing into persistence.

Test 2 — Promotion Requires Independent Support

Input: a candidate persists but has no independent corroboration.

Expected result:

- candidate remains in watch/review state
- promotion is denied or held
- rationale records missing support
- no stronger consequence is authorized

Failure mode caught: repetition alone becoming commitment.

Test 3 — Contradiction Blocks Promotion

Input: a candidate meets persistence threshold but receives strong contradictory evidence.

Expected result:

- promotion is blocked
- candidate is decayed, revised, or quarantined
- contradiction evidence is logged
- audit record preserves decision path

Failure mode caught: promotion continuing after support becomes unstable.

Test 4 — Governance Emits an Audit Artifact

Input: any promoted transition.

Expected result:

- transition emits audit record
- predicate vector is stored
- provenance chain is attached
- decision is replayable

Failure mode caught: silent state mutation.

9. What Is Not Being Claimed

This note makes a narrow architectural claim only.

It does not claim:

- a universal theory of intelligence, consciousness, emotion, or meaning;
- certified safety, clinical validity, regulatory compliance, or deployment readiness;
- that weak signals are decisive or should always escalate;
- that one promotion predicate, threshold set, or schema fits every domain.

The claim is limited to boundary discipline: systems handling weak signals need explicit separation between transient evidence and durable consequence.

10. Durable Contribution

The durable contribution of the PUTMAN / Spanda sequence is a compact specification for preventing boundary collapse.

The main failure pattern is repeated across domains:

- observation becomes truth
- salience becomes persistence
- interpretation becomes authority
- association becomes commitment
- recall becomes authorization
- weak evidence becomes action before it has earned scope

The stack prevents that failure by separating the stages and requiring governed promotion.

A weak signal may matter. It may shape attention. It may request more evidence. It may remain alive as a candidate. But stronger consequence requires support, provenance, stability, and an auditable decision.

11. Conclusion

This consolidation note reduces the PUTMAN / Spanda bridge sequence to one implementable pattern: **Notice → Interpret → Shape → Govern**. The practical artifact is a weak-signal state machine, a promotion predicate template, a minimal data model, and a test checklist for boundary discipline.

The point is not to make weak signals look stronger than they are. The point is to keep them from being either ignored or prematurely hardened. The architecture gives uncertainty somewhere to live before it becomes memory, authority, escalation, or action.

That is the center of gravity of the project.